Gastrointestinal (GI) OSCE Cases

Question and answer review by exam year, 2018 to 2025

Key: Drug names **Critical point** ★ Repeated case Answers follow ACG, AGA, ASPEN and ACG 2024 H. pylori guidelines.

2025

Case 1: Crohn's Disease with TPN Calculations ★ Repeated case, also in 2022–2023

Case details: 21-year-old man, 60 kg, mild terminal ileal Crohn's disease on Budesonide 9 mg PO. The weight changes between scenarios: always re-check the weight on screen before calculating.

Question	Answer
Q1. Minimum and maximum duration of Budesonide?	• Induction course: 9 mg once daily for **8 weeks** (label), up to about 12 weeks maximum • Then taper: 6 mg daily for 2 weeks (label); many taper 6 mg then 3 mg for 2 weeks each • Not for maintenance: no benefit beyond 3–6 months; repeat courses only for recurrence • Why chosen: ileocecal release with high first-pass metabolism (fewer steroid effects) • Non-pharm: stop smoking (worsens Crohn's), nutrition and vitamin D/calcium, vaccines before immunosuppression • Monitor: CDAI/HBI symptoms at 4–8 weeks, CRP, fecal calprotectin, CBC, weight; steroid effects (glucose, BP, mood, bone density)
Q2. Deterioration; Infliximab will be started. What do you screen?	• **Tuberculosis** (IGRA or TST, chest X-ray if positive) and **hepatitis B** (HBsAg, anti-HBc total, anti-HBs) • Hepatitis C, HIV, varicella status; baseline CBC, LFTs • Review: demyelinating disease, NYHA III–IV heart failure, active infection, malignancy • Vaccines (inactivated) before starting; no live vaccines • Infliximab 5 mg/kg IV at weeks 0, 2, 6, then every 8 weeks; combination with Azathioprine lowers antibody formation
Q3. Tapering of Budesonide?	• 9 mg → 6 mg daily × 2 weeks → (3 mg × 2 weeks) → stop • Do not stop abruptly after a prolonged course: adrenal suppression is possible
Q4. TPN at 30 kcal/kg: total calories?	• 60 kg × 30 = **1800 kcal/day**
Q5. Protein 1.2 g/kg: what are the non-protein calories?	• Protein: 60 × 1.2 = 72 g/day = 72 × 4 = **288 kcal** • Non-protein calories = 1800 − 288 = **1512 kcal** (Dextrose 3.4 kcal/g and lipid 10 kcal/g of a 20% emulsion)
Q6. Screen shows Dextrose 9% at 80 mL/h: grams per day and calories?	• Volume = 80 × 24 = 1920 mL/day • Dextrose = 1920 × 0.09 = **172.8 g/day** • Calories = 172.8 × 3.4 = **587.5 kcal**
Q7. If the patient got 295 g Dextrose/day, what is the GIR?	• GIR (mg/kg/min) = grams × 1000 ÷ weight ÷ 1440 • 295,000 ÷ 60 ÷ 1440 = **3.4 mg/kg/min** (safe limit 4–5; at 70 kg it would be 2.9)
Q8. Potassium phosphate 20 mmol (1 mL = 3 mmol phosphate and 4.4 mEq K): what volume and how much K?	• Volume = 20 ÷ 3 = **6.7 mL** • Potassium = 6.7 × 4.4 = **29.3 mEq** • Count this potassium when adding KCl; check K, Phos, Mg before and during infusion
Also be ready for (exam add-ons)	• **Goals:** induce and maintain steroid-free remission, mucosal healing, prevent complications and surgery, maintain nutrition and growth • **Non-pharm:** stop smoking, nutrition (exclusive enteral nutrition), vitamin D, calcium, bone health, vaccines, avoid NSAIDs • **First-line:** mild ileocecal: Budesonide 9 mg daily × 8 weeks; moderate–severe: Prednisone 40 mg with taper or early biologic (Infliximab 5 mg/kg at 0, 2, 6 weeks then q8 weeks) • **Pregnancy:** aim for remission before conception; continue anti-TNF and Azathioprine; Mesalamine acceptable; short Prednisone courses; avoid Methotrexate (stop ≥3 months before), JAK inhibitors and Thalidomide • **Renal/dialysis:** Mesalamine nephrotoxicity (monitor SCr); Methotrexate avoid CrCl <30; Azathioprine reduce; PN electrolytes restricted • **Liver:** Budesonide systemic exposure rises in Child-Pugh B/C (avoid); Azathioprine and Methotrexate hepatotoxic • **HF:** Infliximab avoided in NYHA III–IV • **GI (PUD/GERD):** add PPI if steroids with NSAID, anticoagulant or prior ulcer • **Counseling:** Budesonide in the morning, swallow whole, no grapefruit juice; steroid effects; Infliximab: infusion reaction, infections, TB symptoms; do not stop abruptly

2024

Case 1: Crohn's Disease (Mild Ileal) with TPN Calculations ★ Repeated case, also in 2025, 2022–2023

Case details: Man with Crohn's disease, weight 60 kg, mild ileal lesion. Later severe deterioration and TPN planned.

Question	Answer
Q1. What is the recommended agent? Dose, dosage form, duration?	• Mild ileocecal Crohn's: **Budesonide extended-release capsule 9 mg once daily in the morning (swallow whole), 8 weeks** (up to 12), then taper 6 mg × 2 weeks (optional) • Avoid grapefruit juice (CYP3A4); not for maintenance
Q2. Severe deterioration: what will you give, and what will you check?	• Admit; steroids: Methylprednisolone 40–60 mg IV daily (or Prednisone 40–60 mg PO), and early biologic (Infliximab 5 mg/kg at weeks 0, 2, 6) when moderate-to-severe • Check: CRP/ESR, CBC, albumin, electrolytes; stool C. difficile and cultures; CT or MR enterography (obstruction, abscess, fistula); TB (IGRA), hepatitis B, HIV, varicella before biologics; surgical review • Nutrition support (TPN if obstruction or not tolerating enteral)
Q3. TPN at 30 kcal/kg: total calories?	• 60 × 30 = **1800 kcal/day**
Q4. Protein 0.8 g/kg: non-protein calories?	• 0.8 × 60 = 48 g × 4 = 192 kcal; non-protein = 1800 − 192 = **1608 kcal** (ASPEN usual 1–1.5 g/kg in an IBD flare)
Q5. Dextrose 11% at 70 mL/h: GIR and calories from Dextrose?	• Volume 70 × 24 = 1680 mL; Dextrose = 1680 × 0.11 = **184.8 g/day** • GIR = 184,800 mg ÷ 60 kg ÷ 1440 min = **2.1 mg/kg/min** (limit 4–5) • Calories = 184.8 × 3.4 = **628 kcal**
Q6. Potassium phosphate 15 mmol (1 mL = 3 mmol phosphate and 4.4 mEq K): volume and K?	• Volume = 15 ÷ 3 = **5 mL**; potassium = 5 × 4.4 = **22 mEq** • Count this K in the total K of the bag; check K, Phos, Mg before and during PN
Also be ready for (exam add-ons)	• **Goals:** induce and maintain steroid-free remission, mucosal healing, prevent complications and surgery, maintain nutrition and growth • **Non-pharm:** stop smoking, nutrition (exclusive enteral nutrition), vitamin D, calcium, bone health, vaccines, avoid NSAIDs • **First-line:** mild ileocecal: Budesonide 9 mg daily × 8 weeks; moderate–severe: Prednisone 40 mg with taper or early biologic (Infliximab 5 mg/kg at 0, 2, 6 weeks then q8 weeks) • **Pregnancy:** aim for remission before conception; continue anti-TNF and Azathioprine; Mesalamine acceptable; short Prednisone courses; avoid Methotrexate (stop ≥3 months before), JAK inhibitors and Thalidomide • **Renal/dialysis:** Mesalamine nephrotoxicity (monitor SCr); Methotrexate avoid CrCl <30; Azathioprine reduce; PN electrolytes restricted • **Liver:** Budesonide systemic exposure rises in Child-Pugh B/C (avoid); Azathioprine and Methotrexate hepatotoxic • **HF:** Infliximab avoided in NYHA III–IV • **GI (PUD/GERD):** add PPI if steroids with NSAID, anticoagulant or prior ulcer • **Counseling:** Budesonide in the morning, swallow whole, no grapefruit juice; steroid effects; Infliximab: infusion reaction, infections, TB symptoms; do not stop abruptly

2022–2023

Case 1: Crohn's Disease with TPN (Failed Azathioprine) ★ Repeated case, also in 2025

Case details: 50 kg patient with Crohn's disease who failed Azathioprine. Now cannot tolerate oral feeding and TPN is planned. Screen shows low Mg, K and Phosphate and low albumin. (A patient who failed Azathioprine is not "mild".)

Question	Answer
Q1. What do you recommend for the disease?	• !! Mesalamine is not effective in Crohn's disease: do not choose it (AGA/ACG recommend against) • Azathioprine failure = moderate–severe disease → **biologic:** Infliximab 5 mg/kg (or Adalimumab 160/80/40 mg, Ustekinumab, Vedolizumab, Risankizumab), ideally with an immunomodulator • Bridge: Prednisone 40 mg daily with taper (or Budesonide for ileocecal disease) • Screen TB/HBV first; nutrition support • Non-pharm: smoking cessation, nutrition support (enteral/parenteral while NPO), vaccines (inactivated), vitamin D and calcium • Monitor: CDAI, CRP, fecal calprotectin, CBC, albumin at baseline and 8–12 weeks; endoscopy/MRI at 6–12 months; infections, TB/HBV before biologics, infusion reactions, drug trough/antibodies if loss of response
Question	**Answer**
Q2. What are the monitoring parameters of Azathioprine, long term, and the major long-term adverse effect?	• Baseline TPMT/NUDT15 genotype or activity (dose 2–3 mg/kg only if normal) • CBC with differential weekly × 4, then monthly × 3, then every 3 months; LFTs, bilirubin, renal function; lipase if abdominal pain

Question	Answer
	• Early: pancreatitis (first weeks), leukopenia, hepatotoxicity, nausea • **Long term: malignancy** (lymphoma including hepatosplenic T-cell lymphoma in young men, non-melanoma skin cancer): sun protection and yearly skin check • Interaction: Allopurinol (reduce Azathioprine dose by about 75%)
Q3. TPN planned. What do you correct first (low Mg, K, Phosphate)?	• Correct **all electrolytes** before and during the first days of feeding • Replace by level and renal function (IV Potassium phosphate, Magnesium sulfate, Potassium chloride)
Q4. What do you calculate before correcting?	• Assess malnutrition and refeeding risk (weight loss, intake, ASPEN/GLIM criteria): albumin is an inflammation marker, not a nutrition marker • Correct calcium for albumin: Ca (corrected) = measured Ca + 0.8 × (4 − albumin) • Check renal function and urine output before replacing K, Mg and Phos
Q5. Why correct them?	• To avoid **refeeding syndrome** (hypophosphatemia, hypokalemia, hypomagnesemia, fluid overload, thiamine deficiency) • Give Thiamine 100 mg IV before Dextrose; start at about half of goal calories • Monitor: K, Mg, Phos every 12 h for the first 3 days, then daily until stable; glucose every 4–6 h; weight, fluid balance, edema, heart rate
Q6. Total calories at 30 kcal/kg?	• 30 × 50 kg = **1500 kcal/day**
Q7. Protein 0.8 g/kg: protein calories?	• 0.8 × 50 = 40 g × 4 kcal/g = **160 kcal** • Note: 0.8 g/kg is low for malnourished Crohn's; ASPEN usual 1–1.5 g/kg
Q8. Calories from other macronutrients?	• 1500 − 160 = **1340 kcal** (Dextrose + lipid)
Q9. Final protein concentration if the rate is 70 mL/h?	• Volume = 70 × 24 = 1680 mL/day • 40 g ÷ 1680 mL = **2.38%** (about 2.4 g/100 mL)
Also be ready for (exam add-ons)	• **Goals:** control the disease after thiopurine failure, restore nutrition, prevent complications • **Non-pharm:** smoking cessation, enteral/parenteral nutrition, vitamin D, vaccines • **First-line:** biologic (Infliximab 5 mg/kg or Adalimumab 160/80/40 mg) ± immunomodulator; Prednisone 40 mg bridge • **Pregnancy:** anti-TNF may continue; Azathioprine acceptable; avoid Methotrexate and JAK inhibitors • **Renal/dialysis:** PN: restrict K, Phos and Mg; adjust thiopurine in renal impairment • **Liver:** avoid hepatotoxic thiopurine/Methotrexate in decompensated disease • **Counseling:** TB/HBV screening results, infection warning signs, no live vaccines; refeeding warning signs (palpitations, weakness)

Case 2: Crohn's Disease with Perianal Fistula

> **Case details:** Moderate to severe intestinal Crohn's with perianal fistula and severe weight loss. Cannot tolerate oral or NG feeding. Failed Azathioprine.

Question	Answer
Q1. TPN at 30 kcal/kg/day including 70 g protein: calories from protein?	• 70 g × 4 kcal/g = **280 kcal**
Q2. What will you check before starting TPN?	• Electrolytes (K, Mg, Phos) and **refeeding risk**; triglycerides, glucose, LFTs, renal function • Thiamine before Dextrose; start low and advance
Q3. TG rose to 4.8 mmol/L. What will you do and how adjust TPN?	• TG 4.8 mmol/L = about 425 mg/dL: above the 4.5 mmol/L (400 mg/dL) limit • **Hold lipid emulsion** until TG <4.5 mmol/L, then resume at a lower dose (≤1 g/kg/day, slow rate) • Keep a minimum lipid (about 0.5 g/kg or twice weekly) to prevent essential fatty acid deficiency if prolonged • Check for other causes (pancreatitis, Propofol, steroids, sepsis); avoid overfeeding Dextrose; consider fish-oil emulsion
Q4. How will you treat his intestinal Crohn's?	• Infliximab 5 mg/kg IV at weeks 0, 2, 6, then every 8 weeks (best evidence for fistulizing disease); screen TB/HBV first • Add an immunomodulator (Azathioprine failed: consider Methotrexate) to reduce antibodies • Drain any abscess (MRI pelvis, seton); avoid corticosteroids with abscess or fistula • Non-pharm: nutrition support, stop smoking, perianal hygiene and sitz baths, surgical drainage/seton for abscess • Monitor: fistula drainage, CDAI, CRP, calprotectin, pelvic MRI at 3–6 months; infection, infusion reactions, hepatotoxicity, TB/HBV before and yearly
Question	**Answer**
Q5. What are the 3 antibiotics for the perianal fistula, and for how long?	• Metronidazole 500 mg PO TID (10–20 mg/kg/day) • Ciprofloxacin 500 mg PO q12h • Levofloxacin 500–750 mg PO daily (alternative to Ciprofloxacin)

Question	Answer
	• Duration: 4–8 weeks (up to 3 months); watch for Metronidazole neuropathy, tendon and QT effects of quinolones • Monitor: neuropathy (Metronidazole), tendon pain and QT (quinolones), C. difficile, glucose, LFTs; fistula response at 4–8 weeks
Also be ready for (exam add-ons)	• **Goals:** close the fistula, control luminal disease, prevent abscess and sepsis, restore nutrition • **Non-pharm:** perianal hygiene and sitz baths, surgical drainage/seton, nutrition support, stop smoking • **First-line:** Infliximab 5 mg/kg (weeks 0, 2, 6, then q8 weeks) + Metronidazole 500 mg TID and Ciprofloxacin 500 mg BID for 4–8 weeks • **Pregnancy:** avoid Ciprofloxacin and Levofloxacin; Metronidazole acceptable after the first trimester; Infliximab acceptable • **Renal/dialysis:** Ciprofloxacin: CrCl 30–50: 250–500 mg q12h; <30: q18–24h; Metronidazole no adjustment • **Liver:** Metronidazole reduce dose in severe hepatic impairment • **Counseling:** Metronidazole: metallic taste, no alcohol, numbness (neuropathy); Ciprofloxacin: tendon pain, separate from calcium/iron/antacids by 2 h, sun protection

Case 3: Fulminant Ulcerative Colitis ★ Repeated case, also in 2021, 2018–2019

Case details: Hgb <7 and bloody diarrhea more than 10 times per day. Patient on oral and rectal Mesalamine (2021 and 2018–2019 versions: also low K and Mg, high CRP/ESR, C. difficile negative).

Question	Answer
Q1. What is the diagnosis and severity?	• **Acute severe/fulminant UC:** ≥6 bloody stools/day (here >10) plus toxicity: HR >90, T >37.8 °C, Hgb <10.5, ESR >30 (CRP >30) • Rule out C. difficile and CMV; abdominal X-ray for megacolon/perforation
Q2. How do you treat this patient?	• **Admit:** IV fluids, correct K and Mg, transfuse if Hgb <7, VTE prophylaxis (Enoxaparin) despite bleeding • Methylprednisolone 60 mg IV daily (or Hydrocortisone 100 mg IV q8h) for 3–5 days • Assess response on day 3 (stools >8/day, or 3–8 with CRP >45 mg/L = failure) • Failure: rescue with Infliximab 5 mg/kg (accelerated 10 mg/kg possible) or Cyclosporine 2 mg/kg/day IV; surgery (colectomy) if no response or toxic megacolon • Avoid: opioids, anticholinergics, Loperamide, NSAIDs (risk of megacolon) • Non-pharm: admit, bowel rest or low-residue diet, correct fluids and electrolytes, nutrition support, early surgical consultation • Monitor: stool frequency and blood daily, HR, temperature, abdominal exam and X-ray, CRP, Hgb, K, Mg, albumin; day-3 response; steroid side effects (glucose, BP); Cyclosporine levels/Mg/BP/SCr
Q3. What do you check before rescue therapy?	• **Hepatitis B** (HBsAg, anti-HBc) and **tuberculosis** (IGRA); HIV, HCV, varicella; CMV and C. difficile testing • For Cyclosporine: Mg, cholesterol, renal function, BP; start Azathioprine to bridge
Q4. How do you maintain remission?	• Thiopurine: Azathioprine 2–2.5 mg/kg/day (start 50 mg and titrate) after TPMT/NUDT15 • Or Infliximab/Vedolizumab/Ustekinumab maintenance after response • Mesalamine adds little after steroid-requiring severe disease
Q5. Mesalamine counseling (oral and rectal)?	• Oral 2.4–4.8 g/day; rectal 1 g daily at bedtime (enema: lie on the left side, retain 8 hours) • Swallow tablets whole; monitor SCr and urinalysis at baseline, 3–6 months, then yearly • Report rash, fever, flank pain, worsening diarrhea (hypersensitivity, nephritis, pancreatitis)
Also be ready for (exam add-ons)	• **Goals:** rapid remission, avoid colectomy and complications (megacolon, perforation), correct anemia and electrolytes • **Non-pharm:** admit, IV fluids, correct K and Mg, nutrition, VTE prophylaxis, stop NSAIDs/opioids/antimotility drugs, early surgical review • **First-line:** Methylprednisolone 60 mg IV daily (or Hydrocortisone 100 mg q8h); rescue Infliximab 5 mg/kg or Cyclosporine 2 mg/kg/day • **Pregnancy:** still treat severe UC: steroids, Infliximab and Cyclosporine are acceptable; avoid Methotrexate and Tofacitinib; Mesalamine acceptable (Folic acid 2 mg/day) • **Renal/dialysis:** Cyclosporine nephrotoxic (check levels, SCr, Mg); Mesalamine monitor SCr • **Liver:** Cyclosporine and thiopurines hepatotoxic; check LFTs • **HF/HTN:** Cyclosporine raises BP • **Counseling:** Mesalamine: swallow whole, rectal dosing at bedtime and retain 8 h; thiopurine: infection signs; steroid effects

Case 4: Peptic Ulcer Disease and GERD (PPI Durations)

Case details: Peptic ulcer case with GI symptoms.

Question	Answer
Q1. Diagnosis and management with PPI?	• Uncomplicated duodenal ulcer: PPI once daily × **4 weeks** (8 if large or not healed); gastric ulcer: once daily × **8 weeks** • H. pylori: 14-day eradication regimen, then continue the PPI to finish 4 weeks (duodenal) or 8 weeks (gastric) • NSAID-related: stop the NSAID; PPI daily 4–8 weeks (continue PPI if NSAID cannot be stopped) • Bleeding ulcer: high-dose IV PPI 80 mg bolus then 8 mg/h (or 40 mg IV BID) × 72 h, then oral BID × 14 days, then daily • GERD: once-daily PPI 4–8 weeks (30–60 min before breakfast) • Note: some exam answer keys list once-daily 8 weeks for uncomplicated duodenal ulcer; the guideline duration is 4 weeks • Non-pharm: stop NSAIDs, smoking and alcohol; dietary triggers; stress reduction; small meals • Monitor: symptom relief in 1–2 weeks, alarm features (bleeding, weight loss, vomiting), Hgb, H. pylori test of cure ≥4 weeks after therapy, repeat endoscopy for gastric ulcer at 8–12 weeks
Q2. What was the risk factor and what do you counsel?	• GERD: avoid trigger foods (fatty, spicy, caffeine, chocolate, mint, citrus, carbonated drinks), no meals 2–3 h before bed • Weight loss if overweight; stop smoking and alcohol; raise the head of the bed • Take the PPI 30–60 minutes before the first meal
Also be ready for (exam add-ons)	• **Goals:** relieve symptoms, heal the ulcer or esophagitis, eradicate H. pylori, prevent recurrence and complications • **Non-pharm:** stop NSAIDs, smoking and alcohol; avoid trigger foods; no meals 2–3 h before bed; head-of-bed elevation; weight loss • **First-line:** PPI standard dose daily (Omeprazole 20 mg); H. pylori: bismuth quadruple × 14 days; bleeding: IV PPI 80 mg then 8 mg/h • **Pregnancy:** GERD: lifestyle → antacid/alginate/Sucralfate → Famotidine → PPI; defer H. pylori treatment; avoid Bismuth, Tetracycline, Metronidazole in the first trimester • **Renal/dialysis:** PPI no change; Famotidine adjust; avoid Magnesium/Aluminum antacids and Sucralfate in CKD • **Liver:** PPI lower dose in severe disease • **HF:** avoid sodium-rich antacids and NSAIDs • **Counseling:** PPI 30–60 min before breakfast; long-term risks (Mg, B12, fracture, C. difficile); Clopidogrel: prefer Pantoprazole

2021

Case 1: Crohn's Disease and Ulcerative Colitis (Differences, Azathioprine, Counseling) ★ Repeated case, also in 2020, 2018–2019

Case details: Crohn's case (C. difficile negative) with plan to start Azathioprine; also the UC differences question.

Question	Answer
Q1. How to manage and what to test before rescue therapy?	• Induction by severity: mild ileocecal Budesonide 9 mg; moderate–severe Prednisone 40 mg with taper over 8 weeks, then a biologic (anti-TNF, Ustekinumab, Vedolizumab, Risankizumab) ± immunomodulator • Tests before biologics/immunomodulators: TB (IGRA), hepatitis B, HIV, TPMT/NUDT15, varicella, CBC, LFTs • Non-pharm: smoking cessation, nutrition (small frequent low-residue meals during flares), vitamin D and calcium, vaccines before immunosuppression • Monitor: symptoms, CRP, fecal calprotectin, CBC, albumin; endoscopy at 6–12 months; TB/HBV before biologics
Q2. What is the difference between UC and Crohn's?	• **UC:** rectum upward, continuous, mucosa/submucosa only, crypt abscesses, bloody diarrhea, p-ANCA more often • **Crohn's:** any site from mouth to anus (terminal ileum), skip lesions, **transmural**, granulomas, fistulas and strictures, perianal disease, ASCA more often • Inflammatory markers: CRP and fecal calprotectin in both; CRP usually rises more in Crohn's
Q3. Acute phase and maintenance, with doses and counseling?	• Acute: as in Q1; flare of Crohn's: Prednisone 40 mg daily, taper 5–10 mg weekly • Maintenance: Azathioprine 2–2.5 mg/kg/day; or Methotrexate 25 mg SC/IM weekly + Folic acid 1 mg; or the biologic that induced remission • Counseling: take Azathioprine with food, takes 8–12 weeks to work, do not stop suddenly, report fever or sore throat, jaundice, abdominal pain, use sunscreen, no live vaccines

Question	Answer
Q4. Azathioprine: monitoring and counseling?	• CBC weekly × 4, monthly × 3, then every 3 months; LFTs, lipase if pain • Allopurinol interaction; pregnancy planning discussion • Check before starting: TPMT/NUDT15, viral hepatitis, TB
Also be ready for (exam add-ons)	• **Goals:** induce and maintain remission, mucosal healing, avoid steroid dependence and surgery • **Non-pharm:** smoking cessation (Crohn's), nutrition, vitamin D and calcium, vaccines before immunosuppression • **First-line:** Crohn's flare: Prednisone 40 mg taper or Budesonide 9 mg; maintenance Azathioprine 2–2.5 mg/kg; UC mild–moderate: Mesalamine 2.4–4.8 g oral ± 1 g rectal • **Pregnancy:** Azathioprine, anti-TNF and Mesalamine acceptable; avoid Methotrexate, Tofacitinib, Thalidomide • **Renal/dialysis:** reduce Azathioprine in renal impairment; monitor SCr with Mesalamine • **Liver:** avoid Methotrexate; monitor LFTs on Azathioprine • **Counseling:** Azathioprine with food, 8–12 weeks to work, report fever, sore throat, jaundice, sunscreen, no live vaccines, allopurinol interaction

Case 2: GERD (Full Management) ★ Repeated case, also in 2020, 2018–2019

Case details: GERD: full management with doses and duration; lifestyle, PPI safety, monitoring, drugs that worsen reflux and role of antacids.

Question	Answer
Q1. Full management with doses and duration?	• Lifestyle: weight loss, raise the head of the bed, no meals 2–3 h before bed, avoid tight clothes, stop smoking and alcohol, avoid trigger foods • PPI: Omeprazole 20 mg (or Esomeprazole 40 mg, Pantoprazole 40 mg) once daily 30–60 min before breakfast × 8 weeks (4–8) • Partial response: BID PPI or add an H2 blocker at night (Famotidine 20 mg); antacid or alginate PRN • Maintenance: lowest effective dose or on-demand; endoscopy if alarm features, no response, or long-standing symptoms • Goals: relieve symptoms, heal esophagitis, prevent Barrett's esophagus, stricture and bleeding • Monitor: symptoms at 4–8 weeks, alarm features, adherence; long term: Mg and B12 (≥1 year on PPI), bone health
Q2. What is the long-term PPI safety profile?	• Increased C. difficile, enteric infections and microscopic colitis • Low magnesium and B12 (check Mg before starting if ≥1 year); lower iron absorption • Fracture association plausible, not proven; use calcium citrate if supplement needed • Acute interstitial nephritis, CKD association; conflicting data for dementia and pneumonia
Q3. How do you monitor PPIs?	• Symptoms at 4–8 weeks; re-evaluate adherence and other causes if persistent • Labs: magnesium and B12 on long-term therapy • Interactions: Clopidogrel (Omeprazole/Esomeprazole lower activation: prefer Pantoprazole), Rilpivirine (contraindicated), Atazanavir, high-dose Methotrexate, Levothyroxine and Calcium carbonate absorption
Q4. Which drugs can cause GERD symptoms?	• Calcium channel blockers, nitrates, NSAIDs, bisphosphonates, SSRIs, tricyclics, anticholinergics, corticosteroids, Theophylline • Take pills with a full glass of water; avoid Aspirin/Ibuprofen/Naproxen (use Acetaminophen)
Q5. What is the role of antacids?	• On-demand relief of mild symptoms (<1/week); onset within 5 minutes, effect 30–60 min • Examples: Calcium carbonate (Tums), Aluminum/Magnesium hydroxide, Alginic acid (Gaviscon), Sodium bicarbonate • Cautions: sodium load in heart failure; aluminum and magnesium in kidney failure; avoid sodium bicarbonate in pregnancy • Interactions: separate from bisphosphonates, tetracyclines, iron, Levothyroxine by 2–4 hours • Side effects: aluminum constipation, magnesium diarrhea, calcium hypercalcemia/stones
Also be ready for (exam add-ons)	• **Goals:** relieve symptoms, heal esophagitis, prevent Barrett's esophagus, strictures and bleeding • **Non-pharm:** weight loss, head-of-bed elevation, no meals 2–3 h before bed, avoid triggers (fat, caffeine, chocolate, mint, alcohol), stop smoking, loose clothing • **First-line:** Omeprazole 20 mg (or Esomeprazole 40 mg, Pantoprazole 40 mg) daily 30–60 min before breakfast × 8 weeks; BID for partial response; bedtime Famotidine 20 mg • **Pregnancy:** lifestyle → antacid or alginate → Famotidine → PPI (Pantoprazole or Lansoprazole); avoid sodium bicarbonate antacids • **Renal/dialysis:** Famotidine halve the dose if CrCl <50; avoid Magnesium and Aluminum antacids • **Liver:** PPI lower dose in severe disease; PPI raises SBP and HE risk in cirrhosis • **HF:** avoid sodium-rich antacids and alginates • **Counseling:** timing, long-term effects (Mg, B12, fracture, C. difficile), drugs that worsen reflux, antacid separation

2020

Case 1: H. pylori Peptic Ulcer Disease (Stations 6) ★ Repeated 3 times in 2020 (station, detailed, repeated)

Case details: Woman with weight loss and epigastric pain worse with meals; back pain treated with Ibuprofen. H. pylori positive.

Question	Answer
Q1. What are the 4 diagnostic tests (invasive and non-invasive)?	• Non-invasive: **urea breath test** and **stool antigen** (serology only shows exposure) • Invasive (endoscopy with biopsy): rapid urease test, histology, culture/PCR with susceptibility • PPIs stop ≥2 weeks, antibiotics and bismuth ≥4 weeks before UBT, stool antigen and biopsy tests
Q2. What are the risk factors and medication causes?	• H. pylori; NSAIDs and Aspirin; smoking, alcohol; severe illness; Zollinger-Ellison (rare) • Drugs: NSAIDs, Aspirin, corticosteroids (with NSAIDs), SSRIs, anticoagulants (bleeding), Potassium chloride, bisphosphonates, Doxycycline
Q3. How will you treat, and what are two more regimens?	• First line (ACG 2024): **bismuth quadruple × 14 days** (PPI BID + Bismuth + Tetracycline 500 mg QID + Metronidazole 500 mg TID–QID) • Clarithromycin triple (PPI BID + Clarithromycin 500 mg BID + Amoxicillin 1 g BID × 14 days) only if clarithromycin resistance <15% and no prior macrolide • Rifabutin triple, concomitant therapy, or Vonoprazan 20 mg BID + Amoxicillin 1 g TID × 14 days (dual); penicillin allergy: bismuth quadruple • Stop the NSAID; continue PPI to heal the ulcer (4–8 weeks) • Non-pharm: stop smoking and alcohol, avoid NSAIDs, adherence counseling • Monitor: adherence and side effects at 1–2 weeks; test of cure ≥4 weeks after finishing; ulcer healing (endoscopy for gastric ulcer)
Q4. Which test confirms eradication, and when?	• UBT or stool antigen ≥4 weeks after finishing antibiotics and ≥2 weeks off PPI • Required for ulcer, MALT lymphoma or persistent symptoms
Q5. Back pain on Ibuprofen: should he get Celecoxib? Why?	• Avoid NSAIDs while the ulcer heals; first-line analgesic Acetaminophen up to 3 g/day or topical NSAID, physiotherapy • If an NSAID is unavoidable: **Celecoxib 100–200 mg daily plus PPI**, after H. pylori eradication; lower GI risk than non-selective NSAIDs • Caution: Celecoxib has cardiovascular risk and interferes with Aspirin cardioprotection • Monitor: bleeding signs, BP, renal function
Also be ready for (exam add-ons)	• **Goals:** eradicate H. pylori, heal the ulcer, prevent recurrence and bleeding, stop the NSAID • **Non-pharm:** stop NSAIDs, smoking and alcohol; stress and meal habits; test of cure • **First-line:** bismuth quadruple × 14 days (PPI BID + Bismuth + Tetracycline 500 mg QID + Metronidazole 500 mg TID–QID); alternatives: clarithromycin triple, rifabutin triple, Vonoprazan dual • **Pregnancy:** defer eradication until after delivery; analgesia with Acetaminophen • **Renal/dialysis:** avoid Bismuth subsalicylate and Tetracycline in advanced CKD • **Liver:** Metronidazole dose reduction in Child-Pugh C • **HF:** avoid NSAIDs; Celecoxib has cardiovascular risk • **Counseling:** black stool, no alcohol with Metronidazole, Tetracycline apart from dairy, sun protection; complete 14 days

Case 2: Crohn's Disease Flare and Azathioprine ★ Repeated case, also in 2021, 2018–2019

Case details: Crohn's flare; team requests Azathioprine.

Question	Answer
Q1. How do you treat a flare?	• Mild ileocecal: Budesonide 9 mg daily; moderate–severe: Prednisone 40 mg daily, taper 5–10 mg weekly over 8 weeks • Early biologic ± immunomodulator for moderate–severe or high-risk disease (Infliximab + Azathioprine = SONIC) • Non-pharm: stop smoking, nutrition support (exclusive enteral nutrition in some), hydration, avoid NSAIDs • Monitor: CDAI, CRP, calprotectin, CBC, weight; steroid effects (glucose, BP, mood), response at 2–4 weeks; TB/HBV before biologics
Q2. What are the maintenance options?	• Azathioprine 2–2.5 mg/kg/day; Methotrexate 25 mg weekly SC/IM + Folic acid 1 mg daily • Biologics: Infliximab, Adalimumab, Vedolizumab, Ustekinumab, Risankizumab • Do not use steroids for maintenance; stop smoking
Q3. Azathioprine: dose, monitoring and education?	• 2–3 mg/kg/day after normal TPMT/NUDT15 (start 50 mg, increase) • CBC and LFTs on the schedule above; allopurinol interaction; skin cancer and lymphoma risk • Education: take with food, effect after 8–12 weeks, infection warning signs, sunscreen, no live vaccines

Question	Answer
Q4. Compare Crohn's and UC in diagnosis and histopathology.	• Crohn's: segmental, transmural inflammation, **non-caseating granulomas**, fissures, strictures/fistulas, ileal disease, ASCA positive • UC: continuous from the rectum, mucosal and submucosal, crypt abscesses and architecture distortion, p-ANCA positive • Endoscopy with biopsies, MRI/CT enterography for small bowel, fecal calprotectin
Also be ready for (exam add-ons)	• **Goals:** control the flare, steroid-free maintenance, prevent complications • **Non-pharm:** stop smoking, nutrition support, vaccines, avoid NSAIDs • **First-line:** Prednisone 40 mg daily with a taper over 8 weeks; Azathioprine 2–2.5 mg/kg; biologic if moderate–severe • **Pregnancy:** Azathioprine and anti-TNF continue; avoid Methotrexate • **Renal/dialysis:** Azathioprine dose reduction; Liver: monitor LFTs • **Counseling:** Azathioprine counseling points (infection, sunscreen, no live vaccines, allopurinol interaction)

2018–2019

Case 1: Ulcerative Colitis on Oral and Rectal Mesalamine (Severity and Management)

> **Case details:** Patient on Mesalazine oral and rectal with 10–15 bloody diarrhea, abdominal distension, high CRP and ESR.

Question	Answer
Q1. What is the stage and on what basis?	• **Severe/fulminant UC** (Truelove-Witts): ≥6 bloody stools/day with HR >90, T >37.8, Hgb <10.5, ESR >30 • Mild: <4 stools, little blood, normal ESR; moderate: 4–6 stools, minimal toxicity
Q2. What is your management (acute and chronic)?	• Acute: admit, IV steroids (Methylprednisolone 60 mg IV daily), fluids, electrolytes, VTE prophylaxis, exclude C. difficile • Rescue on day 3 failure: Infliximab or Cyclosporine; colectomy if toxic or non-responsive • Chronic: Azathioprine 2–2.5 mg/kg or a biologic (Vedolizumab, Infliximab); continue Mesalamine for mild disease only • Non-pharm: admission, fluids/electrolytes, nutrition support, avoid NSAIDs/opioids/antimotility drugs • Monitor: stool count, blood, HR, temperature, CRP, Hgb, K, Mg, albumin daily; day-3 steroid response; colonoscopy for dysplasia after 8 years
Q3. What will you monitor?	• Stool frequency, blood, HR, temperature, abdominal exam, CRP, Hgb, K, Mg, albumin • Steroids: glucose, BP, mood; Cyclosporine: levels, Mg, BP, SCr
Q4. Which medications should you avoid in such a severe case?	• NSAIDs, opioids, anticholinergics, Loperamide and other antimotility agents (toxic megacolon); avoid unnecessary antibiotics (C. difficile)
Q5. What is the indicator to start steroids?	• Moderate–severe activity, or **no response to optimal oral plus rectal Mesalamine** after 2–4 weeks • Severe disease: start IV steroids immediately without waiting
Also be ready for (exam add-ons)	• **Goals:** control severe activity, avoid colectomy, maintain remission • **Non-pharm:** admit, fluids and electrolytes, nutrition, VTE prophylaxis, avoid NSAIDs/opioids/antimotility drugs • **First-line:** Methylprednisolone 60 mg IV daily × 3–5 days; rescue Infliximab 5 mg/kg or Cyclosporine; maintenance Azathioprine 2–2.5 mg/kg or a biologic • **Pregnancy:** steroids, Infliximab and Azathioprine are acceptable • **Renal/dialysis:** Cyclosporine nephrotoxic; Liver: monitor LFTs • **Counseling:** Mesalamine adherence, steroid effects, infection signs

SOE Bank (year not stated)

Case 1: NSAID-Induced Gastric Ulcer on Low-dose Aspirin (Previous STEMI)

Case details: 60-year-old woman with intermittent abdominal pain worse with food; endoscopy shows a gastric ulcer. Takes Ibuprofen for back pain and low-dose Aspirin (history of STEMI).

Question	Answer
Q1. What is the goal of therapy?	• Relieve symptoms, heal the ulcer, prevent complications (bleeding, perforation) and recurrence, keep cardioprotection
Q2. What is the diagnosis?	• **NSAID-induced gastric ulcer** (look for H. pylori as a co-factor)
Q3. What is the management?	• Stop Ibuprofen (it also blocks the antiplatelet effect of Aspirin); test for H. pylori (UBT, stool antigen or biopsy) and eradicate if positive (bismuth quadruple × 14 days) • **PPI:** Pantoprazole 40 mg daily (or Omeprazole 20–40 mg) for **8 weeks** for a gastric ulcer (4 weeks if duodenal); continue the PPI long term while on Aspirin • **Continue Aspirin 81 mg** (secondary prevention after STEMI); only hold if bleeding, restart as soon as hemostasis is secure • Repeat endoscopy with biopsy at 8–12 weeks (gastric ulcer: exclude malignancy)
Q4. The doctor suggests switching to Celecoxib for back pain: what do you think?	• First choice: **Acetaminophen up to 3 g/day**, topical Diclofenac, heat and physiotherapy • If an NSAID is unavoidable (high GI and high CV risk): the answer key prefers **Naproxen 250–500 mg BID + PPI** (lowest CV risk among NSAIDs); Celecoxib + PPI is the alternative when GI risk dominates, but COX-2 inhibitors carry CV risk and lose their GI advantage with Aspirin • Lowest dose, shortest time, review at 2–4 weeks
Also be ready for (exam add-ons)	• **Goals:** relieve symptoms, heal the ulcer or esophagitis, eradicate H. pylori, prevent recurrence and complications • **Non-pharm:** stop NSAIDs, smoking and alcohol; avoid trigger foods; no meals 2–3 h before bed; head-of-bed elevation; weight loss • **First-line:** PPI standard dose daily (Omeprazole 20 mg); H. pylori: bismuth quadruple × 14 days; bleeding: IV PPI 80 mg then 8 mg/h • **Pregnancy:** GERD: lifestyle → antacid/alginate/Sucralfate → Famotidine → PPI; defer H. pylori treatment; avoid Bismuth, Tetracycline, Metronidazole in the first trimester • **Renal/dialysis:** PPI no change; Famotidine adjust; avoid Magnesium/Aluminum antacids and Sucralfate in CKD • **Liver:** PPI lower dose in severe disease • **HF:** avoid sodium-rich antacids and NSAIDs • **Counseling:** PPI 30–60 min before breakfast; long-term risks (Mg, B12, fracture, C. difficile); Clopidogrel: prefer Pantoprazole